RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty	:	Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title	:	OML 751 Testing of Materials
Year and Branch	:	IV year Civil, EEE and ECE
Topics Covered	:	Unit -II: Mechanical Testing –Impact Test: Charpy
Name of the Laboratory	:	Strength-of-Materials lab
Website Link	:	http://sm-nitk.vlabs.ac.in/exp6/index.html
Date	:	09.09.2020
No.of Students participated	:	63
Assignment given if any (If Yes share the uploaded folder link)	:	-

Demonstration of Impact Test: Charpy

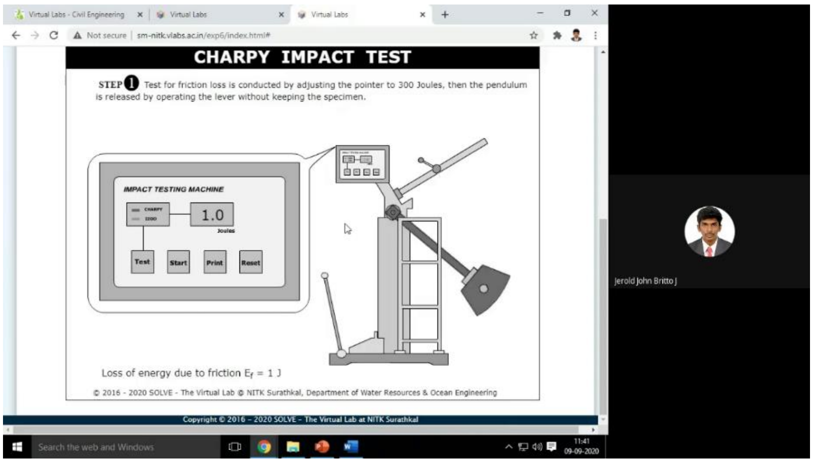
OBJECTIVE

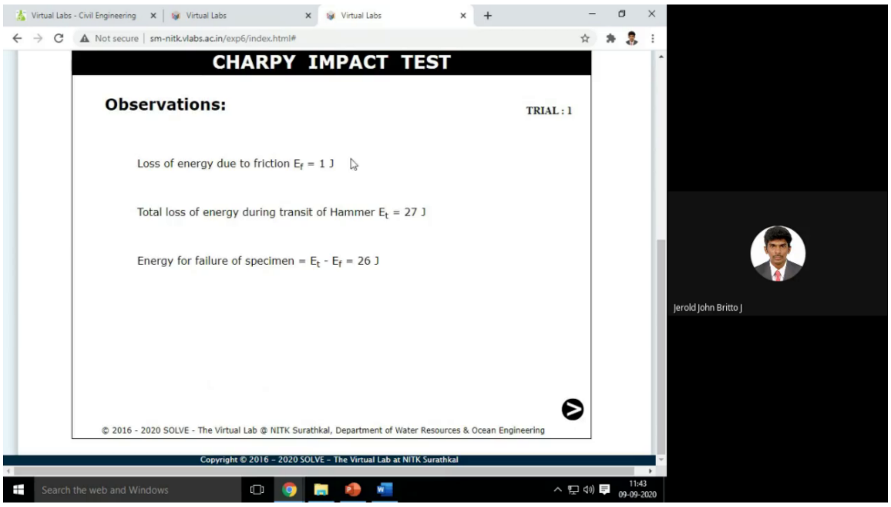
To find the impact resistance of mild steel specimen and cast iron.

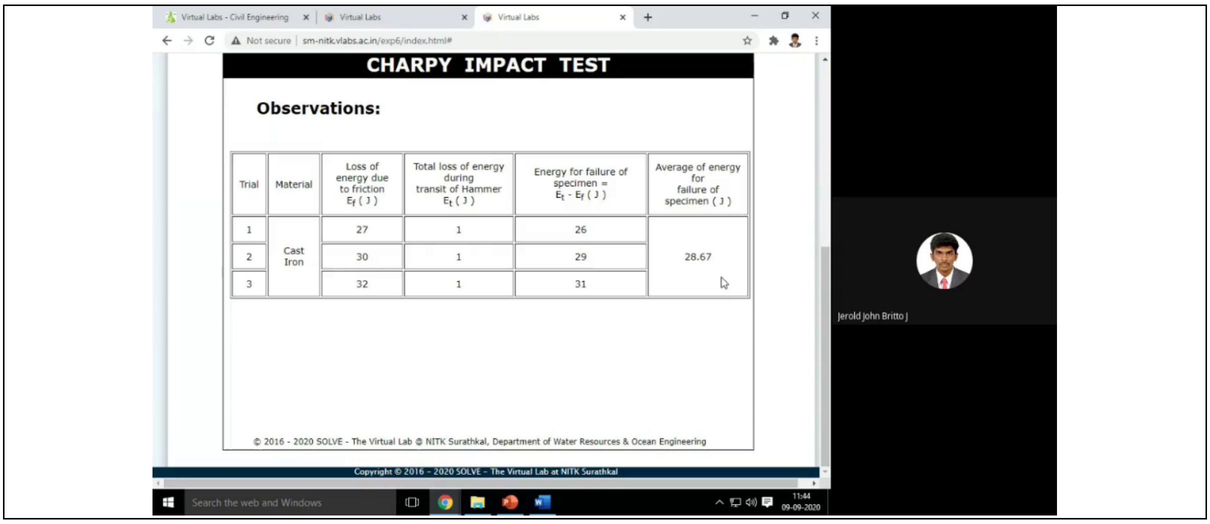
According to Indian Standard the speed of pendulum at the instant of striking shall be 4.5 - 7 m/s and the plane of swing of the striker shall be vertical and within 0.5 mm of the plane midway between the supports. The pendulum can be raised to any desired height and rested at that position. It is supported in the starting position by a catch and can be released by a trigger.

The students can able to describe the toughness mechanism using experimental approach and also understand the Charpy test specimen dimensions, types of notch and experimental step by step procedure.

Course Outcome / Programme Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO8**	**PO9**	**PO10**	**PO12**
CO2 Analyze engineering components using various mechanical testing procedure.	2	2	1	1	2	2	2	3	1







CHARPY IMPACT TEST

Observations:

Trial	Material	Loss of energy due to friction E_f (J)	Total loss of energy during transit of Hammer E_t (J)	Energy for failure of specimen = $E_t - E_f$ (J)	Average of energy for failure of specimen (J)
1	Cast Iron	27	1	26	28.67
2		30	1	29	
3		32	1	31	